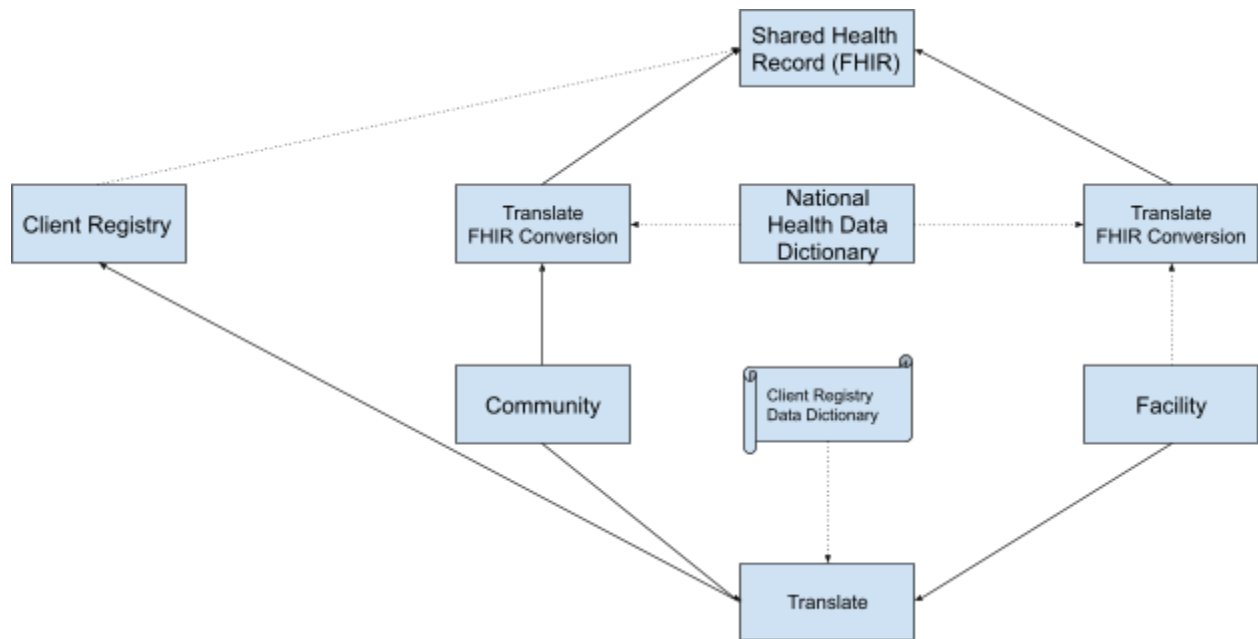


Health Break Integrations

Resources: Link to Design docs & instances

1. [Technical Requirements](#)
2. [FHIR Documentation](#)
3. [National Health Data Dictionary](#)
 - <https://nhdd-api.health.go.ke/swagger/>
 - <https://nhdd-api.health.go.ke/orgs/MOH-KENYA/sources/nhdd/concepts/?limit=10000&includeMappings=true>
4. [Client Registry API](#)
5. [Client Registry Access](#)
6. OpenHIM instance <https://interoperabilitylab.uonbi.ac.ke/openhim/#/>
 - Username: root@openhim.org
 - Password: 1cY6&IWnc0#G
 - API URL: <https://interoperabilitylab.uonbi.ac.ke/api/>
7. eCHIS: <https://chis-training.health.go.ke/>
 - reata_test_cha
 - _ReataCHA1
 - reata_test_chv
 - _ReataCHV1
8. eCHIS v2 config <https://github.com/moh-kenya/config-echis-2.0>
9. [Sample Service Request for Facility to Community](#)

Overview of the systems involved



Integration use cases

- Community <> Client Registry: Status - **OPEN**
- Facility <> Client Registry: Status - **DONE**
- Facility <> Shared Health Record (SHR): Status - **OPEN**
- Community <> Shared Health Record (SHR): Status - **OPEN**

Community <> Facility / Facility <> Community

Questions:

- How are the following coded?
 - Referral requests
 - Successful referrals
 - Missed referrals

Community <> Facility

- Evaluate the **Service Request** FHIR resource
- How do we code the services rendered?
 - At minimum, capture

- What we are referring for
- When the referral is made
-

Facility <> Community

- Coding the Referral Request to community
 - Query the **Encounter Resource** to summarize what happened at the facility
 - Update the **status** in the **Service Request**
- Options from eCHIS
 - Query referral request for status
 - Create a **subscription** to call back eCHIS based on updates to a Service Request
 - For complete service requests, query the encounter resource and summarize what happened.

System Roles

eCHIS

Publish

- New contacts/clients
- Updated contacts/clients
- Referral requests

Subscribe

- Encounters from referrals
- Lost to follow up

Shared Health Record

Publish

- Referral to community
- Encounters
 - Fetch based on referral request status

Subscribe

- Event-based (rather than pull-based)
 - Based on referral request status

Team assignments

Group 1: Community <> Client Registry

Name	Email	Work package assigned
Irene Mhuri*	irene@medic.org	☐ Meet with design team for required fields in the client registry ☐ Update the client registration forms to include necessary fields ☐ Configure outbound push to service ☐ OpenHIM: Build mediator to handle inbound payload
Karim Kanji	karim.karanja@health.go.ke	☒ Provide access to GH repo with prototype service ☐ Update eCHIS contact document with upi ☐ OpenHIM: Build mediator to handle outbound payload
Derick Lung'aho*	derick@medic.org	
Dickson Kirathe	dkirathe@nltp.co.ke	☐ Provide access to Client Registry ☐ OpenHIM: Configure testing environment for OpenHIM to work with eCHIS dev environment
George Bush Odhiambo	georgebush.odhiambo@medtroniclabs.org	

Readiness checklist

Tech	Checklist	ToDos
eCHIS	☒ API Reference	• Map the Client Registry

	documentation ☒ Outbound push documentation ☒ Local environment setup ☐ Data Points: design vs integration cross check ☒ eCHIS Repo access ☐ Dev environment ☐ Sample eCHIS contact documents	Fields and eCHIS contact doc fields 1:1. • Meet with design team for required fields in the client registry • Update the client registration forms with the necessary fields • Configure outbound push to service • Update eCHIS contact document with upi
Client registry	☐ Reference documentation ☐ Sample Registry Payload	
Integration service i.e. OpenHIM		• Build mediator to handle inbound payload • Build mediator to handle outbound payload • Configure testing environment for OpenHIM to work with eCHIS dev environment

Group 2: [Community <> Facility](#)

Name	Email	Work package assigned
Ian Minara	ian.omwenga@health.go.ke	8, 10, 11, 12, 13
Kitsao*	emmanuel@medic.org	1, 3, 4, 9, 12
Faith Jepkoech	fjepkoech@healthit.uonbi.ac.ke	6, 13, 2, 3
Sam K'Olala	sam.o'kolala@health.go.ke	8, 10
Ezra Mungai	ezra.mungai@health.go.ke	7, 8
Eric Maira	ericodemus@gmail.com 0711496978	6, 4
Lewis Kimani	lewiskimani18@gmail.com	5, 10

Readiness checklist

Tech	Checklist	ToDoS
eCHIS	☒ Reference documentation for outbound payload ☒ Complete Documentation ☒ Configuration repository ☒ Data Points: design vs integration cross check ☒ Priority use case identified (ANG) ☒ Dev environment ☐ CU codes from KMCHUL updated ☐ Link facilities updated ☐ CHV codes updated ☐ Outbound payload mapped	1. Update CU details in eCHIS to include CU code from KMCHUL 2. Update CHV Area details in eCHIS to include link facility from KMHFL 3. Update CHV details in eCHIS to include CHV code from CHV registry 4. Configure outbound push module to send data to OpenHIM
OpenHIM	☒ Reference documentation ☒ GHT repo for interoperability ☐ Mediator to handle inbound payload authored ☐ Mediator to handle outbound payload authored ☐ Dev environment User: root@openhim.org Password: (shared here) ☐ Testing environment	5. Build mediator to handle inbound payload (ensure NHDD codes are correctly captured for each referral reason) 6. Build mediator to handle outbound payload 7. Configure testing environment for OpenHIM to work with eCHIS dev environment
FHIR Server	☒ Reference documentation ☒ API ☒ Dev environment ☒ Resource identified and reviewed ☒ Sample resource	8. Service request validation

	mapping ☒ Access granted ☐ Service request validation	
NHDD	☒ Access granted ☒ API ☐ Data points for priority use case mapped to NHDD codes	9. Map data points outlined in the brainstorming doc for the use case to corresponding NHDD code
AfyaKe	☒ Dev environment ☐ SHR consumption for community referrals	10. Consume referral from eCHIS
Kafka	☐ Make it public	11. Make a public Kafka instance
KMHFL	☐ Reference ☐ API ☐ Link facility codes updated for existing deployment (<i>this will involve working with the county team alongside Lwala, for Migori, to hasten the process</i>).	☐
KMCHUL	☐ Reference ☐ API ☐ CU codes for existing deployment updated	☐
CHV registry	☐ Reference ☐ API ☐ CHV details updated	☐
KHIS	☐ Priority matrix done (515) ☐ Access to dev environment granted ☐ Data points mapping done ☐ Datasources to extract MoH 515	12. Build mediator to process inbound payload for MoH 515 13. Build mediator to process inbound payload for MoH 515 14. Harmonize CUs from eCHIS to KHIS

Group 3: Facility <> Community

Name	Email	Work package assigned
Eric Nderitu*		
Eric Ejimba	eejimba@goldindata.org	
Cynthia Chebet	cchebet@healthit.uonbi.ac.ke	
Njuguna Ndung'u	njuguna@medic.org	
Moses Mugambi	moses.mugambi@health.go.ke	
Ambrose Kimaiyo*	akimaiyo@healthit.uonbi.ac.ke	
Ivy Watitwa	iwatitwa@goldindata.org	
Hawi Odhiambo	hawi.odhiambo@health.go.ke	
Sam Kolala	sam.o'kolala@health.go.ke	
Nicholas ingosi	ningosi@healthit.uonbi.ac.ke	
Simon Injiri	sinjiri@healthit.uonbi.ac.ke	
Philip Mwago	mwago@medic.org	

Readiness checklist

Tech	Checklist	ToDoS
AfyaKe	☐ SHR generation for facility to community referrals	1. Add CU details with CU code, linked facilities to AfyaKe 2. Initiate referral request from AfyaKE to FHIR server. Publish referral request to Kafka.
eCHIS	☐ Reference documentation ☐ Data Points: design vs integration cross check ☐ Dev environment	3. Consume referral request from AfyaKE

OpenHIM	☐ Reference documentation ☐ Testing instance	4. Subscribe to referral messages from Kafka. 5. Create a mediator to receive referral messages from FHIR server. 6. Create a mediator to process referral messages. 7. Create a mediator to post referral messages to eCHIS.
Kafka	☐ Reference documentation ☐ Testing instance	8. Publish referral messages from AfyaKe to Kafka.
FHIR Server	☐ Reference documentation ☐ Testing instance and credentials	9. Troubleshoot subscription requests
Grafana		
CHV Registry	☐ Make registry standalone	
Household Registry	☐ Requirements	

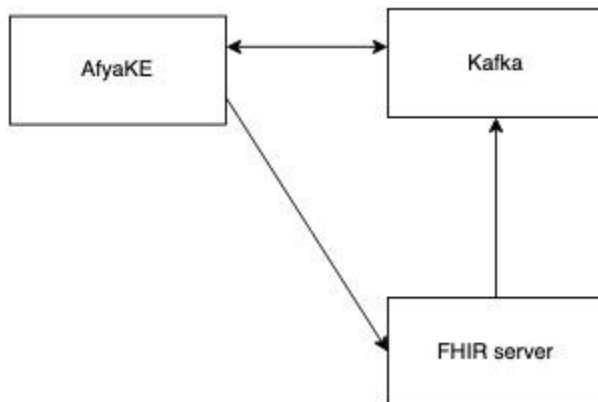
Group 4 Community to KHIS

Name	Email	Work assigned
Oliver Munyao*	omunyao@healthit.uonbi.ac.ke	- Build mediator to process inbound payload for MoH 515 - Build mediator to process outbound payload for MoH 515
Boniface Mugwe	bmugwe@healthit.uonbi.ac.ke	
Kitsao		Get data from database to OpenHIM

Crucial notes

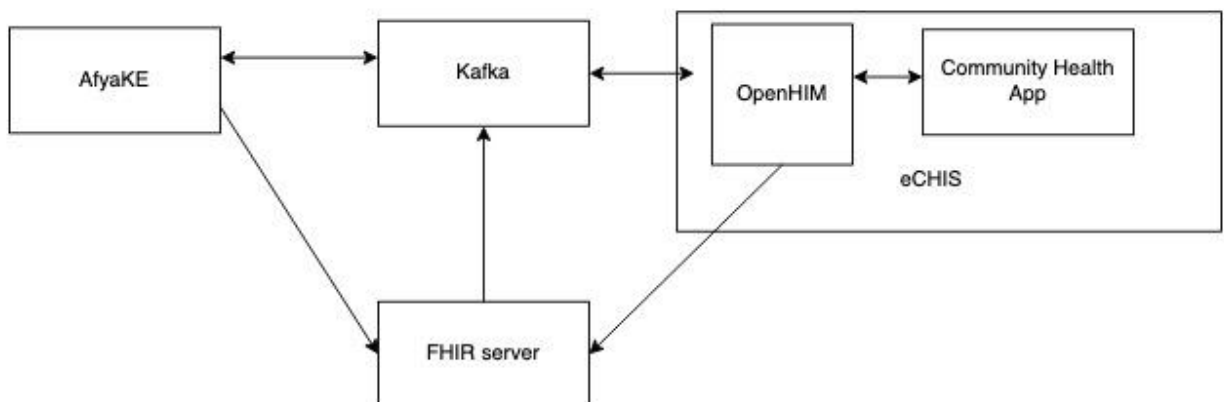
1. Team leads should ensure that each member is on the same page.
2. Team leads have the mandate to drop non contributing group members, bring it to the attention of Dr. Sitienei

Current FHIR implementation with AfyaKE



The proposed architecture for having AfyaKE integrated with FHIR is to have it make calls to the FHIR server directly. The FHIR server can then be configured to publish new messages to Kafka once resources are created. AfyaKE will then subscribe to the Kafka stream/queue and process any resources that seem relevant to it. AfyaKE and Kafka are already talking to each other, but the call to the FHIR server is an integration that will be added at a later date.

Proposed AfyaKE, FHIR, and eCHIS implementation



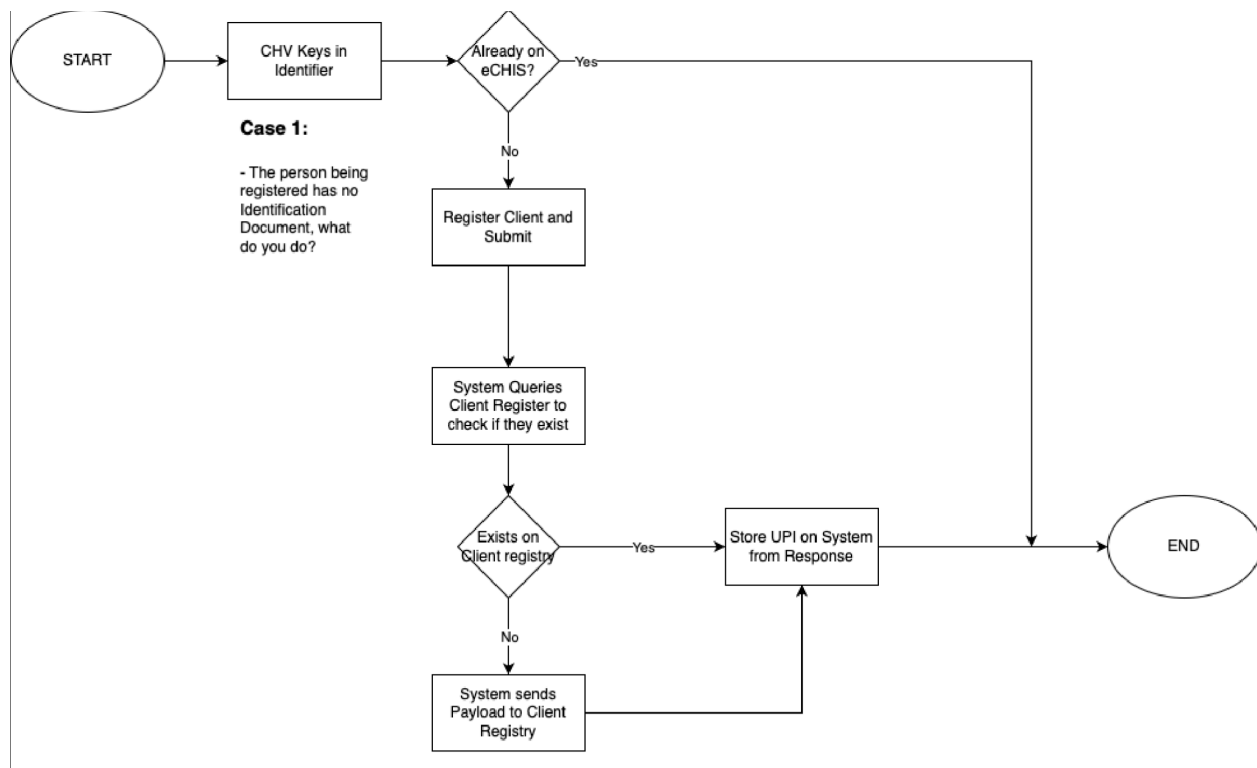
Extending the architecture above. We would include an OpenHIM deployment that would write to FHIR and subscribe to Kafka. All communication to and from the eCHIS system would be through the OpenHIM deployment through dedicated mediators.

eCHIS - KHIS Integration

Client Registry

The following are the required fields in the client registry:

```
{
  "FirstName": "string",
  "LastName": "string",
  "DateOfBirth": "string",
  "Gender": "string",
  "Country": "string",
  "CountyOfBirth": "string",
  "Residence": {
    "County": "string",
    "SubCounty": "string",
    "Village": "string"
  },
  "Identifications": [
    {
      "IdentificationType": "string",
      "IdentificationNumber": "string"
    }
  ],
  "Contact": {
    "PrimaryPhone": "string",
  },
  "NextOfKins": [
    {
      "Name": "string",
      "Relationship": "string",
      "Residence": "string",
      "Contact": {
        "PrimaryPhone": "string",
        "SecondaryPhone": "string",
        "EmailAddress": "string"
      }
    }
  ]
}
```



Sample Service Request for Facility to :

```

{
  "resourceType": "ServiceRequest",
  "status": "active",
  "intent": "order",
  "priority": "routine",
  "code": {
    "coding": [
      {
        "system": "https://nhdd-api.health.go.ke/orgs/MOH-KENYA/sources/nhdd/concepts/30089/names/246275/",
        "code": "30089",
        "display": "Referred to Primary Care"
      }
    ]
  },
  "subject": {
    "reference": "Patient/{{patient_id}}"
  }
}

```

```

"requester": {
  "reference": "Organization/{{facility_id}}"
},
"performer": [
  {
    "reference": "Organization/{{chu_id}}"
  }
],
"note": [
  {
    "text": "clinical instructions"
  },
  {
    "text": "pharmacy instructions"
  },
  {
    "text": "prescription instructions"
  }
]
}

```

Sample Community to Facility eCHIS payload

```

{
  upi: 'MOH0JM2EYYK2',
  name: 'Mahenzo',
  service: 'pregnancy_home_visit',
  screening: {
    _danger_signs: 'vaginal_bleeding fits severe_abdominal_pain',
    _mental_danger_signs: 'lack_of_sleep feeling_of_worthlessness
staying_away_from_people',
    _observed_mental_danger_signs: 'lack_of_hygiene anxiety'
  },
  supportingInfo: {
    _other_danger_sign: '',
    _other_mental_danger_sign: '',
    _other_observed_mental_danger_signs: '',
    _muac_color: 'yellow',
    _has_lost_pregnancy_within_last_month: ''
  }
}

```

```

    _takes_iron_or_folate_supplements: 'yes',
    _is_still_pregnant: 'yes',
    _has_started_anc: '',
    _anc_upto_date: 'yes',
    _has_lost_pregnancy_within_last_month_subsequent_visit: '',
    _pregnancy_test_result: '',
    _is_pregnant: '',
    _is_on_family_planning: '',
    _period_of_pregnancy: '',
    _known_lmp: '',
    _next_anc_visit_date: '2023-04-28'
  },
  reported_date: 1681284129122,
  follow_up_date: '2023-04-15',
  chu_code: '602563',
  chu_name: 'Bamba Community Unit',
  referred_to_facility_code: '11237',
  referred_to_facility_name: 'Bamba Sub-County Hospital'
}

```

Sample Community to Facility Service Request

```

{
  "resourceType": "ServiceRequest",
  "status": "active",
  "intent": "order",
  "category": [
    {
      "coding": [
        {
          "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
          "code": "28414",
          "display": "Patient referred for medical consultation"
        }
      ],
      "text": "Consultation"
    }
  ],
  "priority": "urgent",

```

```
"subject": {
  "reference": "undefined/partners/registry/search/upi/MOH0JM2EYYYYK2",
  "type": "Patient",
  "display": "MOH0JM2EYYYYK2"
},
"occurrencePeriod": {
  "resourceType": "Period",
  "start": "2023-04-12",
  "end": "2023-04-15"
},
"authoredOn": "2023-04-12",
"requester": {
  "reference": "undefined/orgs/MOH-KENYA/sources/KMHFL/concepts//602563",
  "type": "Organization",
  "display": "Bamba Community Unit"
},
"performer": [
  {
    "reference":
"undefined/orgs/MOH-KENYA/sources/KMHFL/concepts//11237",
    "type": "Organization",
    "display": "Bamba Sub-County Hospital"
  }
],
"reasonCode": [
  {
    "coding": [
      {
        "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
        "code": "49168",
        "display": "Vaginal Bleeding"
      },
      {
        "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
        "code": "40356",
        "display": "Fits"
      },
      {
        "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
        "code": "18523",
```

```
      "display": "Severe Abdominal Pain"
    },
    {
      "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
      "code": "29541",
      "display": "Lack of sleep"
    },
    {
      "code": "247892001",
      "display": "Feeling of worthlessness"
    },
    {
      "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
      "code": "29644",
      "display": "Staying away from people"
    },
    {
      "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
      "code": "41667",
      "display": "Lack of hygiene"
    },
    {
      "system": "undefined/orgs/MOH-KENYA/sources/nhdd/concepts/",
      "code": "15111",
      "display": "Anxiety/worried for no apparent reason"
    }
  ]
}
],
"note": [
  {
    "text":
"{\"_muac_color\": \"yellow\", \"_takes_iron_or_folate_supplements\": \"yes\", \"_is_still_pregnant\": \"yes\", \"_anc_upto_date\": \"yes\", \"_next_anc_visit_date\": \"2023-04-28\"}"
  }
]
}
```


Testing Scenarios

Client registry

No.	Scenario	Expected behavior	Comments
1	Client non existing in Registry registered in ECHIS (community)	1. Client created in Registry 2. Client in ECHIS updated with UPI 3. Client accessible in AfyaKE	
2	Client existing in Registry registered in ECHIS (community), without UPI	1. Client in ECHIS updated with UPI 2. Client accessible in AfyaKE	
3	Client existing in Registry registered in ECHIS (community), without UPI	 1. Do nothing 2. Client accessible in AfyaKE 	
4	Client exists in the registry, but biodata does not match e.g. ID matches, but names do not.	1. Create a data mismatch record in CHIS 2. Trigger a CHP client details verification task 3. CHP completes task to update details 4. Outbound Push runs to update UPI	

		in CHIS 5.	
--	--	---------------	--

Community to Facility referral

No.	Scenario	Expected behavior	Comments
1	Service report with referral	1. SHR updated 2. Referral accessible in AfyaKE	
2	Referral missed	1. SHR updated 2. Visit report created in ECHIS 3. Missed visit task to CHV triggered in ECHIS	
3	Referral completed, no community follow up required	1. SHR updated 2. Completed visit report created in ECHIS	
4	Referral completed, community follow up required	1. SHR updated 2. Completed visit report created in ECHIS 3. Follow up task to CHV triggered in ECHIS	

Facility to Community

No.	Scenario	Expected behavior	Comments
1	Service report with referral	1. SHR updated 2. Report created in ECHIS 3. Follow up task to CHV triggered in ECHIS	
2	Referral missed (client not found in the community, CHV did not action task)	1. Visit report created in ECHIS 2. SHR updated 3. Consume in AfyaKE	
3	Referral completed, no facility referral required	1. Visit report created in ECHIS 2. SHR updated 3. Consume in AfyaKE	
4	Referral completed, community follow up required	1. Visit report created in ECHIS 2. SHR updated 3. Consume in AfyaKE	

Sample payload to update existing service request

```
{
  "resourceType": "ServiceRequest",
  "id": "187942e6149-2a07a19b-6217-49e3-8567-657af8da1a84",
  "identifier": [
    {
      "system": "https://afyakenyaapi.health.go.ke/partners/referrals/search/guid/",
      "value": "1fe4e3e5-077e-4d9f-a3c1-f987181f3f73"
    }
  ],
}
```

```

    {
      "system": "https://chis-staging.health.go.ke//medic",
      "value": "1fe4e3e5-077e-4d9f-a3c1-f987181f3f73"
    }
  ],
  "status": "draft",
  "intent": "order",
  "category": [
    {
      "coding": [
        {
          "system": "http://snomed.info/sct",
          "code": "710915002",
          "display": "Referral to community service"
        }
      ],
      "text": "Referral to community service"
    }
  ],
  "priority": "routine",
  "subject": {
    "reference": "https://afyakenyaapi.health.go.ke/partners/registry/search/upi/MOHVDKG18GM58",
    "type": "Patient",
    "display": "MOHVDKG18GM58"
  },
  "occurrencePeriod": {
    "start": "2023-04-18T11:44:40.2278200Z",
    "end": "2023-04-18T11:44:40.2278201Z"
  },
  "authoredOn": "2023-04-18T11:44:40.2278149Z",
  "requester": {
    "reference": "https://nhdd-api.health.go.ke/orgs/MOH-KENYA/sources/KMHFL/concepts/15126",
    "type": "Organization",
    "display": "15126"
  },
  "performer": [
    {
      "reference": "https://nhdd-api.health.go.ke/orgs/MOH-KENYA/sources/KMHFL/concepts/702962",
      "type": "Organization",
      "display": "702962"
    }
  ],

```

```
"note": [  
  {  
    "text": "{\\"health_facility_contact\\":\\"254123098574\\",\\"follow_up_instruction\\":\\"<p>tupo sawa</p>\\"}"  
  },  
  
]  
}
```

Notes from Aug 28, 2023 through Aug 30, 2023 at 67 Airport Hotel

- [CHIS to Client Registry Workflow Sequence and Flow Chart](#)
 - [Raw .drawio file](#)
 - Notes
 - Getting a UPI relies on deterministic details matching using the Levenshtein distance algorithm, implemented by the Client Registry team.
 - Upon matching, the Client Registry validates the same details against the Integrated Population Registration Database System (IPRS) and if found, the client is marked as passed validation, otherwise validation fails.
 - The Client Registry shall eventually have more records than IPRS.
 - For existing contacts in eCHIS, CHPs shall update details to include the minimum required set to get a UPI. We have worked on a task to do so using the client-edit form.
 - For mismatching details, the CHP shall also get a task to verify the details they submitted vs the set from Client Registry before the client gets a UPI.
- [CHIS to SHR Workflow Sequence and Flow Chart](#)
 - [Raw .drawio file](#)
 - Notes
 - There were a few modifications agreed upon for the service requests we create, such as defaulting to a status: active rather than draft.
 - Mapping of attributes for referral (Outbound Push and NHDD) is in progress.
 - One of the arising issues is that we need to listen to all updates to the SHR and cherry pick those for clients in CHIS to create the corresponding records.
 - Please checkout the [referrals ServiceRequest enhancements](#) to configure an updated resource.

Revised Referral Sample Service Request Payload